# Tone in Ixpantepec Nieves Mixtec Word Prosody

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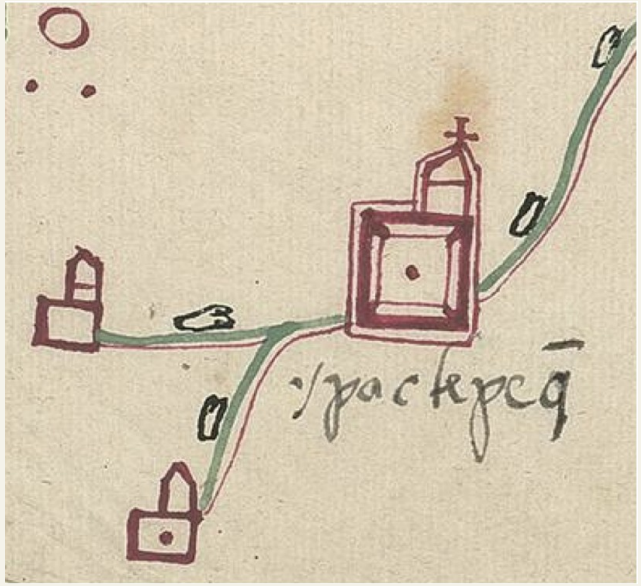
# Objective

Identify key aspects of tone phonology and acoustics in Mixtec of Ixpantepec Nieves (MIN)

- Mixtec languages have
  - complex lexical tone systems
  - word-level stress accent
  - phonemic laryngeal features
- Mixtec languages have closely aligned
  - prosodic structure
  - morphological structure

MIN tone phonology: 1 T (H/M/L) per mora +1
MIN tone acoustics: rightward coarticulation

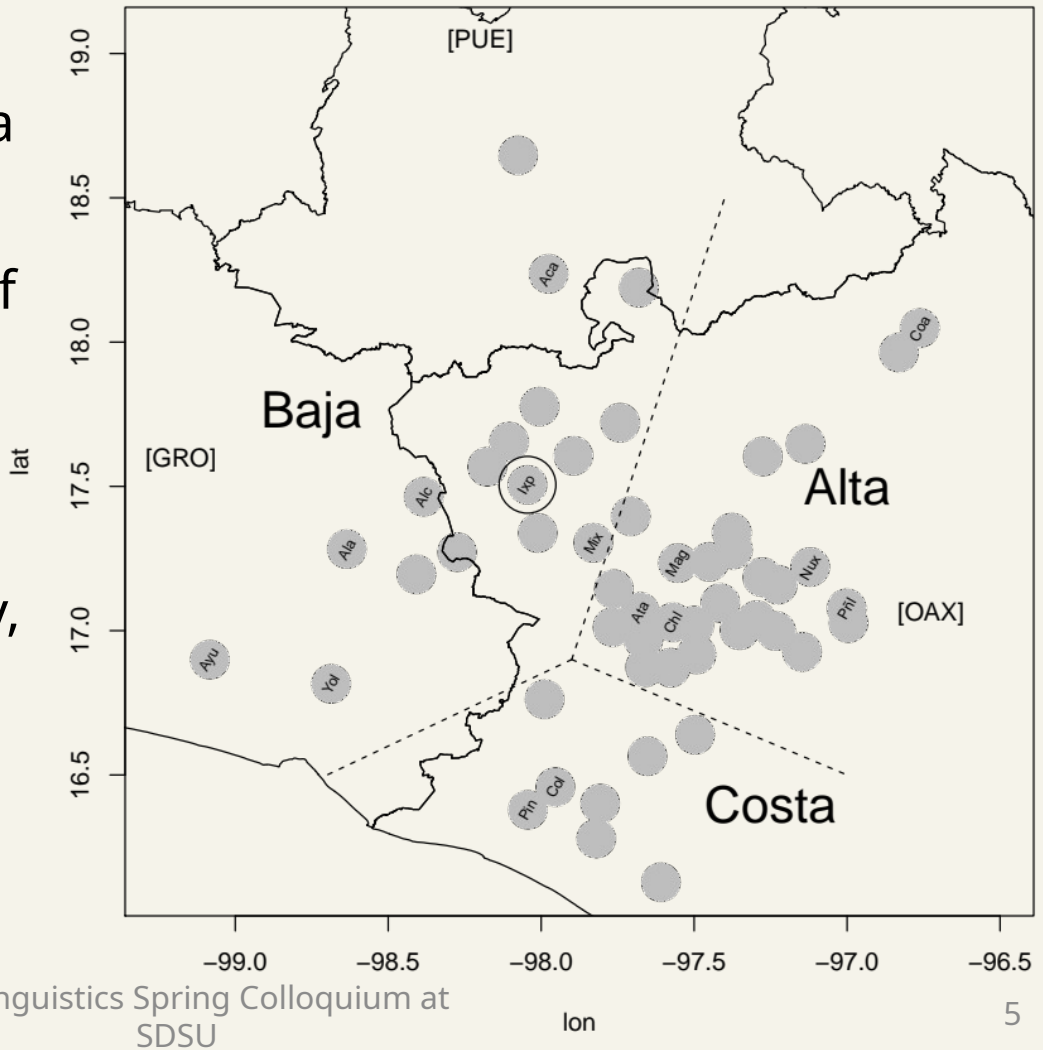
*"El pueblo de Ycpatepec esta dos leguas de Çilacaioapa, … es tierra fragosa y lengua misteca."*
(The town of Ixpantepec is two leagues from Silacayoapan, … the terrain is rough and the language is Mixtec.)
*-Relaciones geográficas de la Diócesis de Tlaxcala*
(Aznar c. 1580/1905:238)



# Language Overview

# Mixtec Geography

- Language continuum in southern Mexico, split into Baja, Alta and Costa regions
- Ixpantepec Nieves (IXP in map): in NW corner of Oaxaca state, on the northern edge of Josserand's (1983) Southern Mixteca Baja dialect group.
- For the tonal phonology, major distinction is central varieties vs. some peripheral ones



# MIN Phonology

- Bimoraic roots
  (plain: CVː, CVCV; glottalized: CVʔV, CVʔCV)
- Monomoraic affixes/clitics (plain: CV)
- Root-initial stress:

(1) [kò.ní(ˈnānā)ɾā]

/kò´ = nì = nana = ɾa/

NEG.RE = PFV = rise = 3MS

- Voiced C: Prenasalized, Nasal, Continuant
- Voiceless C: Stops, Fricatives
- Vowels: Oral, Nasal

# MIN Segments

Oral vowels: /i, e, a, o, u/ +long
Nasal vowels: /ĩ, ã, ũ/, /ĩː, ẽː, ãː, ũː/

| | | labial | coronal | | | dorsal | |
|---|---|---|---|---|---|---|---|
| | | | apico-alveolar | lamino-alveolar | post-alveolar | | lab'ized |
| voiceless | stop | | t | $t^j$ | tʃ | k | $k^w$ |
| | cont | | s | | ʃ | x | |
| voiced | stop | ($^m$b) | $^n$d | $^n$d$^j$ | | ($^n$g) | |
| | cont | v | l | | ʒ | | |
| | nasal | m | n | | ɲ | | |

# Acoustics of Prosody

| Correlate | Word Stress | Phrasal Accent | Tone | Phonation |
|---|---|---|---|---|
| V Duration | ND, SP, ZP | | CM | |
| C Duration | ND | ND | | |
| V Intensity | ND, SP | SW | CM | |
| Low-band spec tilt (H1-H2) | | | | GJ, MZ |
| Mid-band spec tilt (H1-A2) | ND, SP | SW | | GJ, MZ |
| Periodicity (HNR, CPP) | | | | MZ |
| F0 | | EN, SW | MZ | GJ |

MIN
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CM: Mandarin (Shen 1993)
EN: English (Campbell & Beckman 2007)
GJ: Gujarati (Keating et al 2011)
MZ: Mazatec (Garellek & Keating 2011)

ND: Dutch (Cho & McQueen 2005)
SP: Spanish (Ortega-Llebaria & Prieto 2007)
SW: Swedish (Heldner 2003)
ZP: Zapotec (Chavez Peon 2008)

"This observation leads me to the hypothesis that (c) morphemes are those which historically perhaps ended on a tone [L], whatever their present form, and that this low toneme now usually transfers itself to the following morpheme." (Mak 1953:92)

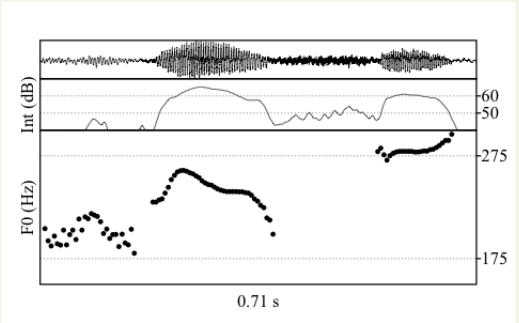
# Tonal Phonology

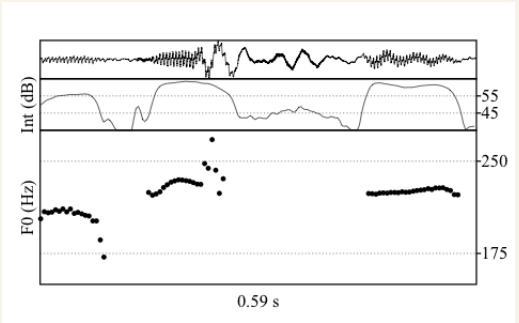
# Tone

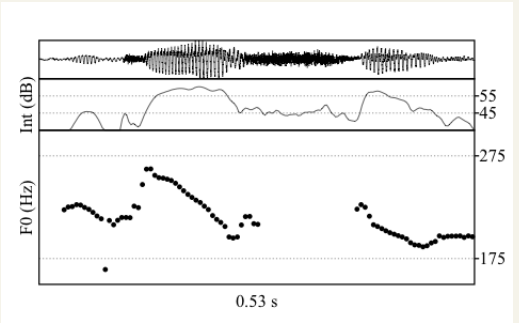
- Tone is L, M or H, with mora as TBU

(12)

| | | |
|---|---|---|
| a. HH: [kíʃí] | b. MM: [kīʃī] | c. LL: [kìʃì] |
| ‘sleepyhead’ | ‘RE.make’ | ‘RE.sleep’ |
| d. MH: [ⁿdʲīʃí] | e. MM: [ⁿdʲīʃī] | f. ML: [ⁿdʲīʃì] |
| ‘blue’ | ‘pulque’ | ‘corn cob’ |



# Plain CVCV Tone Patterns

| | | | |
|---|---|---|---|
| 5.22) | a. H.H: jíkó<br>[xi˥ko˥]<br>/xíkó/<br>‘tall’ | b. M.H: ijá<br>[i˦xa˥]<br>/ixá/<br>‘overmorrow’ | c. L.H: chi̲kí<br>[ʧi˩ki˥]<br>/ʧíkí/<br>‘cactus pear’ |
| 5.23) | a. H.M: sáko<br>[sa˥ko˦]<br>/sáko/<br>‘opossum’ | b. M.M: tyuku<br>[tʲu˦ku˦]<br>/tʲuku/<br>‘again’ | c. L.M: ki̲ki<br>[ki˩ki˦]<br>/kìki/<br>‘sew’ |
| 5.24) | a. H.L: ndyíka̲<br>[ⁿdʲi˥ka˩]<br>/ⁿdʲíkà/<br>‘wide’ | b. M.L: xiko̲<br>[ʃi˦ko˩]<br>/ʃikò/<br>‘smell’ (n.) | c. L.L: yu̲ku̲<br>[ʒu˦ku˩]<br>/ʒùkù/<br>‘grass’ |

# CVCV Floating Tone Patterns

6.28) a. H.L+H: kán̲i̲
[kã˥nĩ˨˩]
/kánì ´/
'hill'

b. H.H+L: kóchí
[ko˥ʧi˥]
/kóʧí `/
'pig'

6.29) a. M.L+H: yuku̲
[ʒu˧ku˨˩]
/ʒūkù´/
'hill'

b. M.H+L: kasí
[ka˧si˥]
/kasí `/
'eat (sweet)'

c. M.M+L: landyi
[la˧ⁿdʲi˧]
/laⁿdʲi`/
'navel'

6.30) a. L.L+H: ji̲ko̲
[xi˧ko˨˩]
/xìkò´/
'neck'

b. L.H+L: ndyu̲xú
[ⁿdʲu˩ʃu˥]
/ᵐdʲùʃú`/
'vain'

# Tone Linking to Enclitics

## Floating +H tones on M-tone enclitic

| (6.32) | a. | nuu̲ na | b. | nii̲ na | c. | yuku̲ na | d. | jiko̲ na |
|---|---|---|---|---|---|---|---|---|
| | | [nũ˧˥ũ˩˧nã˥] | | [nĩ˧˥ĩ˩˧nã˥] | | [ʒu˧˥ku˩˧nã˥] | | [xi˧˥ko˩˧nã˥] |
| | | /nuù´=na`/ | | /nìì´=na`/ | | /ʒūkù´=na`/ | | /xìkò´=na`/ |
| | | face=3P | | salt=3P | | hill=3P | | neck=3P |
| | | 'their faces' | | 'their salt' | | 'their hill' | | 'their necks' |

## Floating +L tones on M-tone enclitic

| (6.33) | a. | léé na | b. | tyi̲ín na | c. | ndyu̲xú na | d. | kóchí na |
|---|---|---|---|---|---|---|---|---|
| | | [le˥˩e˥˩nã˩] | | [tʲĩ˩˥ĩ˥˩nã˩] | | [ⁿdʲu˩˥ʃu˥˩nã˩] | | [ko˥˩ʧi˥˩nã˩] |
| | | /léé`=na`/ | | /tʲìí$_{n}$`=na`/ | | /ⁿdʲùʃú`=na`/ | | /kóʧí`=na`/ |
| | | baby=3P | | mouse=3P | | vain=3P | | hill=3P |
| | | 'their baby' | | 'their mouse' | | 'they're vain' | | 'their hill' |

# Tone Linking to Roots

Floating +L tone displaces H.H root tones

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| (6.65) | a. | laa | léé | b. | kóchí | léé | c. | ndásá | lẹ́é |
| | | [la˧la˧ | le˩e˥] | | [ko˥ʧi˥ | le˩e˥] | | [ⁿda˥sa˥ | le˩e˥] |
| | | /laa` | léé`/ | | /kóʧí` | léé`/ | | /ⁿdásá` | léé`/ |
| | | bird | baby | | pig | baby | | chachalaca | baby |
| | | ‘baby bird’ | | | ‘baby pig’ | | | ‘baby chachalaca’ | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| (6.66) | a. | ilo | píndó | b. | ilo | kúsú | c. | ita kuáán |
| | | [i˧lo˧ | pi˩ⁿdo˥] | | [i˧lo˧ | ku˩su˥] | | [i˧ta˧kʷã˩ã˩] |
| | | /ilo` | píⁿdó`/ | | /ilo` | kúsú/ | | /ita`–kʷááₙ`/ |
| | | rabbit | speckled | | rabbit | white | | flower–yellow |
| | | ‘a speckled rabbit’ | | | ‘a white rabbit’ | | | ‘marigold’ |

# L+H Proclitics

+H tone from counterfactual /kǒ´/ replaces L tone of PFV /nì/

(6.104)

| a. | | | b. | | |
|---|---|---|---|---|---|
| nị | ka'a | yaa | kọ nị | ka'a | yaa |
| [nĩ˩ka˧ʔa˧ | | ja˧a˧] | [ko˩nĩ˥˧ka˧ʔa˧ | | ja˧a˧] |
| /nì=ka²a | | ʒaa/ | /kò´=nì=ka²a | | ʒaa/ |
| PFV=play | | music | NEG.RE=PFV=play | | music |
| 'there was music' | | | 'there wasn't music' | | |

+H tone from negative potential /ùn´/ replaces root tone

(6.108)

| a. | | | b. | | |
|---|---|---|---|---|---|
| kuaku | na | | ụn kuaku | na | |
| [kʷa˧ku˧nã˧] | | | [ũ˩kʷa˥˩ku˧nã˧] | | |
| /kʷaku=na/ | | | /ù$_n$´=kʷaku=na/ | | |
| IR:cry=3P | | | NEG.IR=IR:cry=3P | | |
| 'they will cry' | | | 'they won't cry' | | |

# +L Proclitic

L tone of /nì/ spreading onto M tone prefix

(6.95)

a. ni kakānī rā
[nĩ˧ka˩kã˧nĩ˧ɾa˧]
/nì=ka–kani=ɾa/
PFV=PL–hit=3M
'they hit (something)'

b. ni nandyikọ-ra
[nĩ˧nã˩ⁿdʲi˩ko˩ɾa˩]
/nì=na–ⁿdʲìkò=ɾa/
PFV=REP–follow=3M
'he followed'

c. ni kusíká na
[nĩ˧ku˩si˥ka˧nã˧]
/nì=ku–síká=na`/
PFV=INCH–far=3P
'they went away'

+L tone linking to M tone prefix as only marker of PFV

(6.97)

a. chinúu ndó
[ʧi˩nu˧u˥ⁿdo˧]
/`\ʧi–núu=ⁿdó/
PFV\put–face=2P
'you put it on top'

b. kakani ra
[ka˩kã˧nĩ˧ɾa˧]
/`\ka–kani=ɾa/
PFV\PL–hit=3M
'they hit (something)'

c. nasino ra
[nã˩sĩ˧nõ˩ɾa˩]
/`\na–sinù=ɾa/
PFV\REP–descend=3M
'he went back down'

# +H Proclitic

+H tone replacing M tone of root

| | | | |
|---|---|---|---|
| (6.100) | a. | ndyayi̲ na<br>[ⁿdʲa˧˩ji˧˩nã˩]<br>/ⁿdʲaʒì = na/<br>shake = 3P<br>'they will shake' | b. | ndyáyi̲ na<br>[ⁿdʲa˥ji˧˩nã˩]<br>/´\ⁿdʲaʒì = na/<br>IPFV\shake = 3P<br>'they are shaking' |

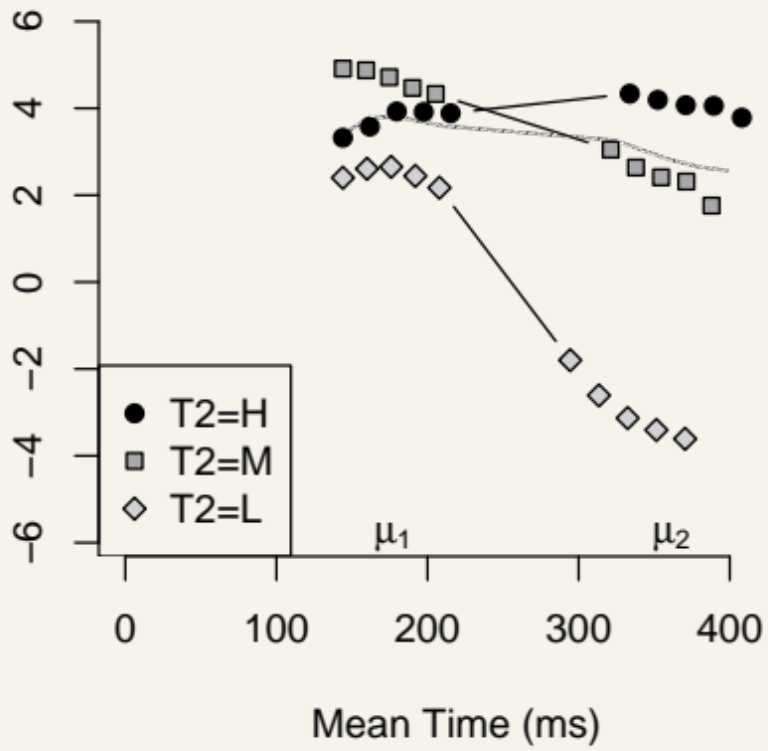
+H tone replacing L tone of root

| | | | |
|---|---|---|---|
| (6.101) | a. | xi̲kó na<br>[ʃi˩ko˥˧na˧]<br>/ʃìkó = na/<br>sell = 3P<br>'they will sell' | b. | xíkó na<br>[ʃi˥ko˥nã˧]<br>/´\xìkó = na/<br>IPFV\sell = 3P<br>'they are selling' |

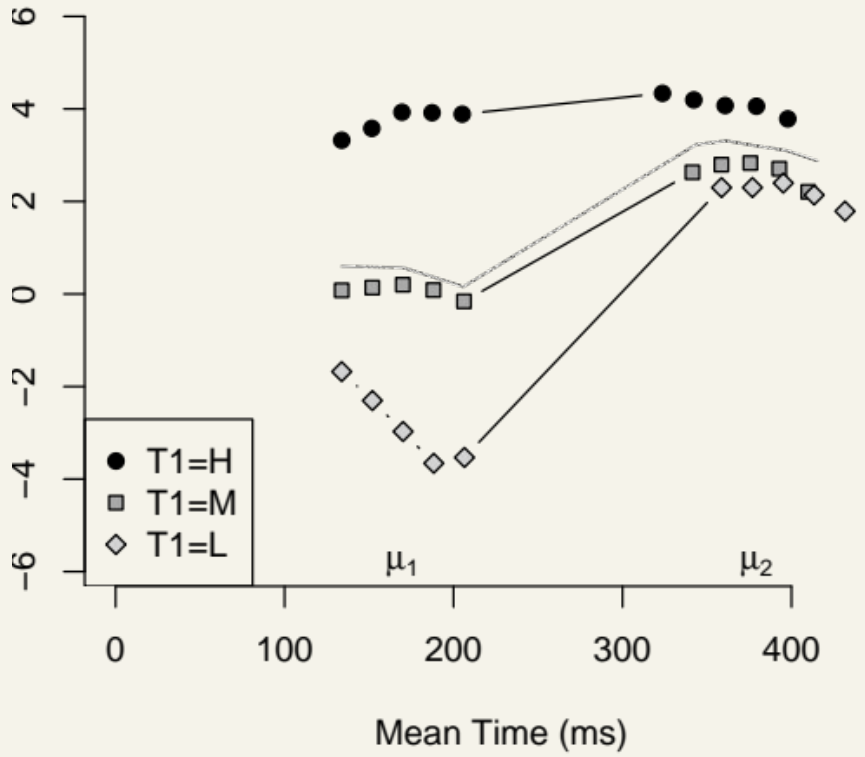
# Tone Acoustics

# H Tone Coarticulation

**Initial H tone**

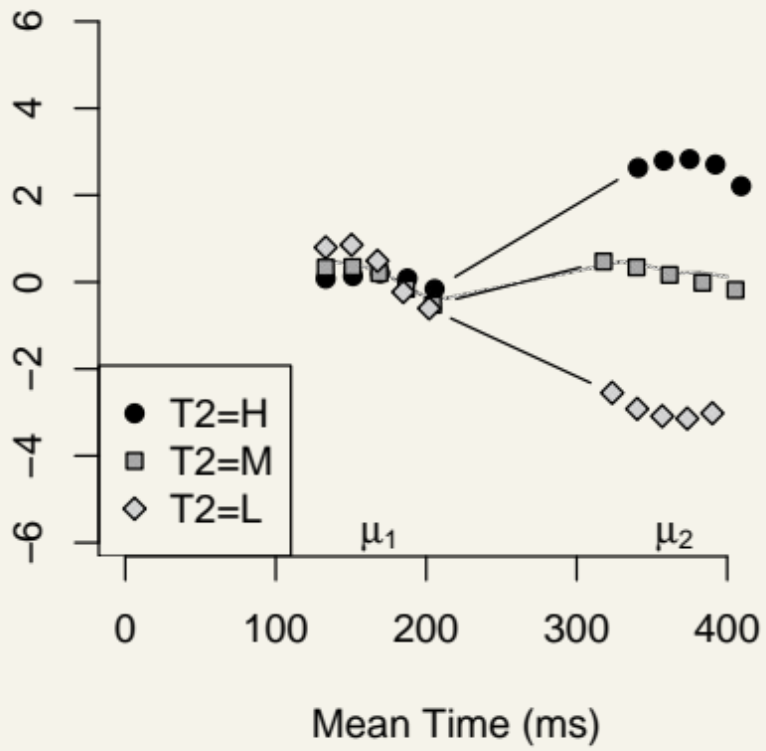



**Final H tone**

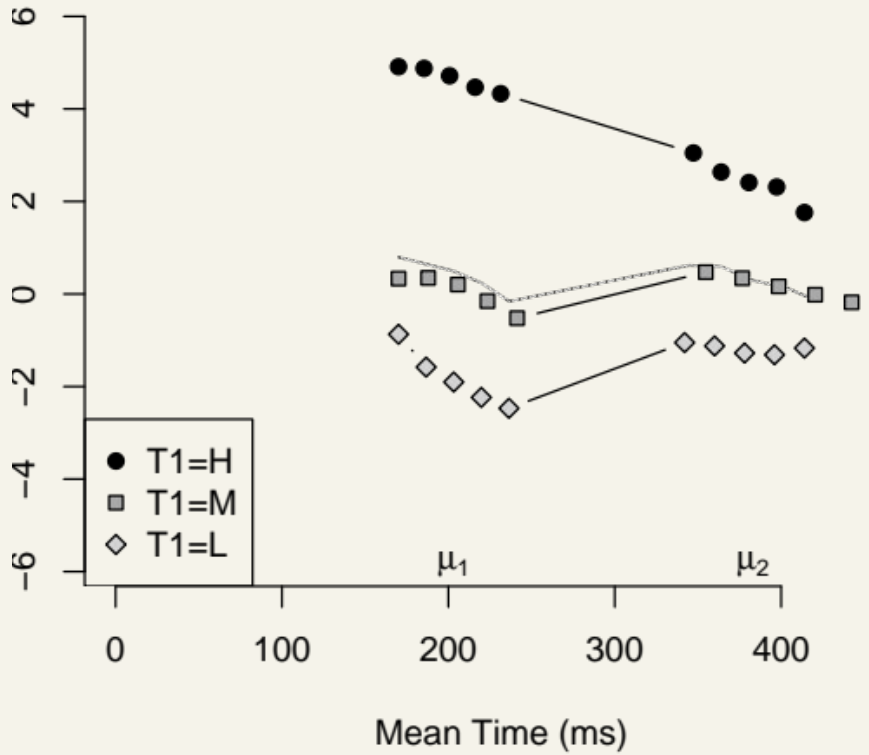


# M Tone Coarticulation

**Initial M tone**

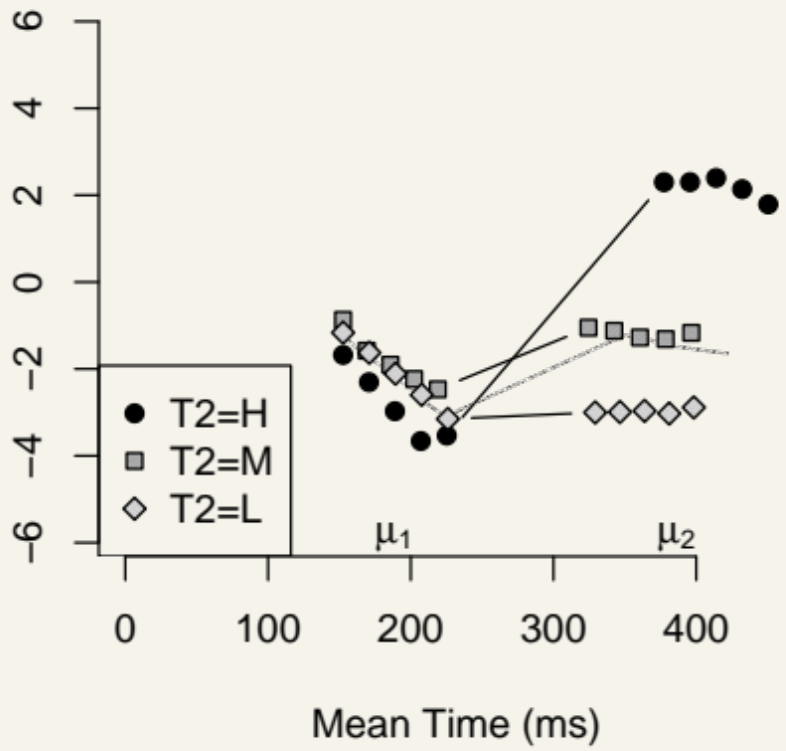



**Final M tone**

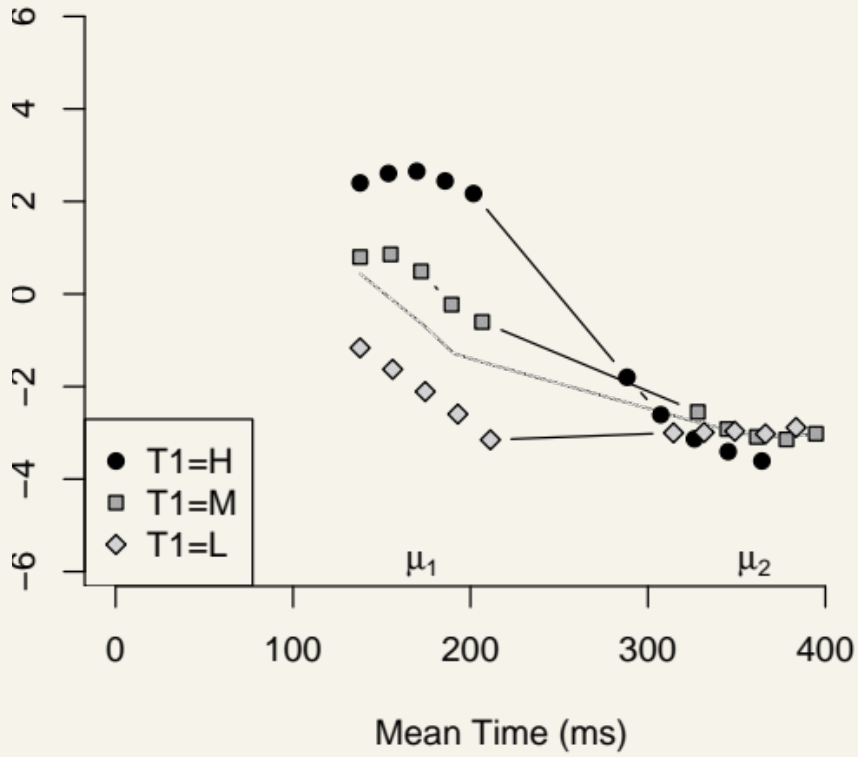


# L Tone Coarticulation

**Initial L tone**




**Final L tone**



# Tone and Phonation Type

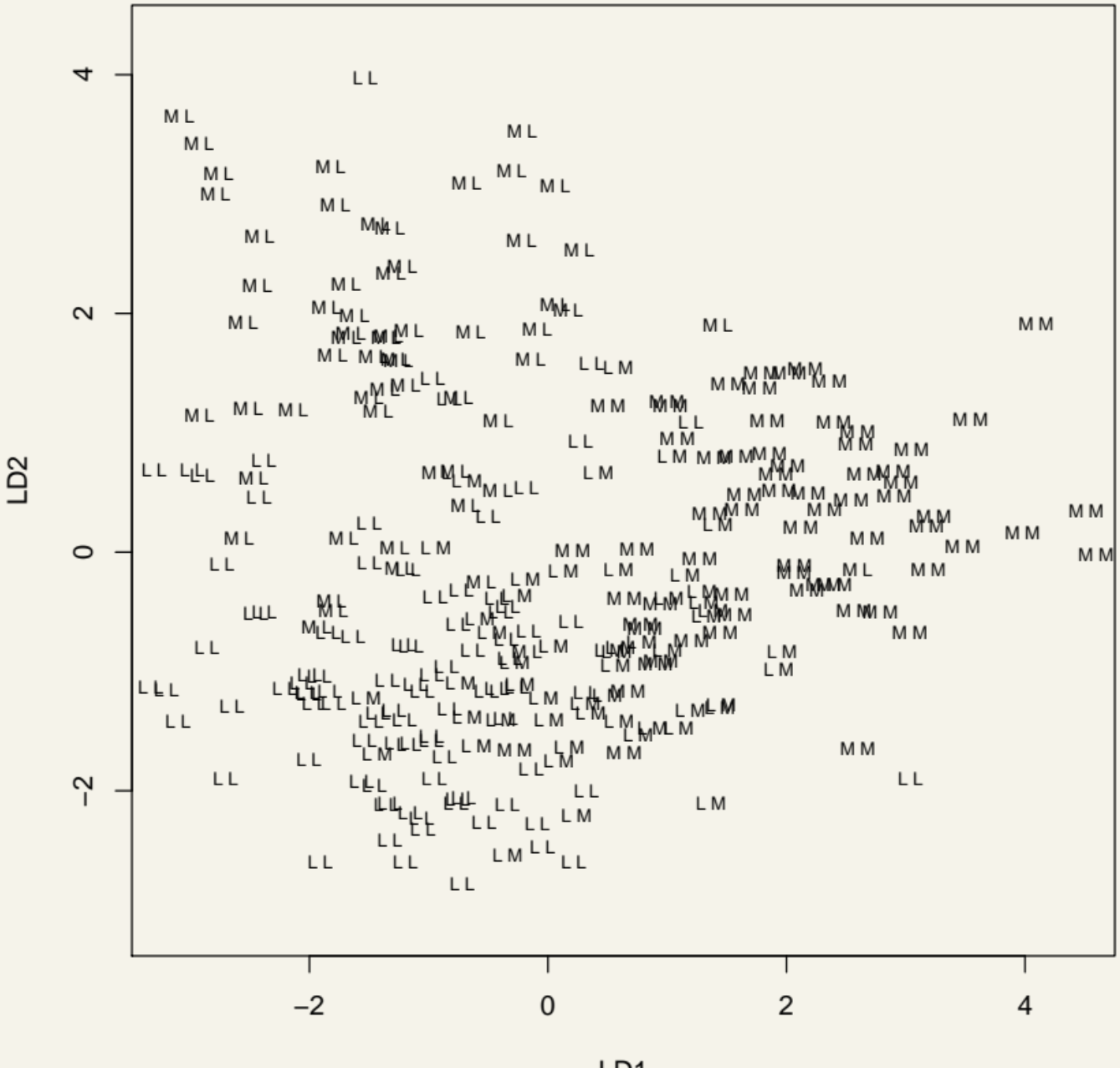
**DC: discriminant coeffs** **SC: structure coeffs**

a.) Glottalization

| | DC | SC |
|---|---|---|
| F0 | 1.036 | 0.422 |
| CPP | -0.877 | -0.830 |
| HNR | -0.628 | -0.706 |
| H1–H2 | -0.341 | -0.273 |
| H1–A2 | 0.033 | -0.455 |
| Int | -0.113 | -0.383 |

b.) Tone

| | $DC_1$ | $SC_1$ |
|---|---|---|
| F0 | 1.694 | 0.915 |
| CPP | 0.496 | 0.401 |
| HNR | 0.005 | 0.443 |
| H1–H2 | 0.255 | 0.650 |
| H1–A2 | 0.139 | 0.245 |
| Int | 0.231 | 0.313 |


LD2
LD1
4
2
0
−2
−2
0
2
4

# Tone and Phonation Type

| | Glottal | Tone $D_1$ | Tone $D_2$ |
|---|---|---|---|
| $F0_i$ | 0.142 | 0.348 | 0.635 |
| $F0_m$ | 0.185 | 0.110 | 0.036 |
| $F0_f$ | 0.549 | -0.049 | 1.047 |
| $CPP_i$ | -0.276 | -0.157 | 0.229 |
| $CPP_m$ | -1.070 | -0.738 | 0.038 |
| $CPP_f$ | 0.001 | -0.299 | 0.031 |
| $HNR_i$ | 0.046 | 0.452 | 0.106 |
| $HNR_m$ | -0.082 | -0.733 | -0.349 |
| $HNR_f$ | -0.268 | -0.508 | 0.111 |

# Conclusions

- Tones H and L phonologically active
- Tone M is less marked
- Morphemes sponsor 1 tone/mora +1
- Coarticulation mirrors phonological tone processes: rightward; H, L > M
- L tone linked to breathy voice, and breathiness spreads leftward

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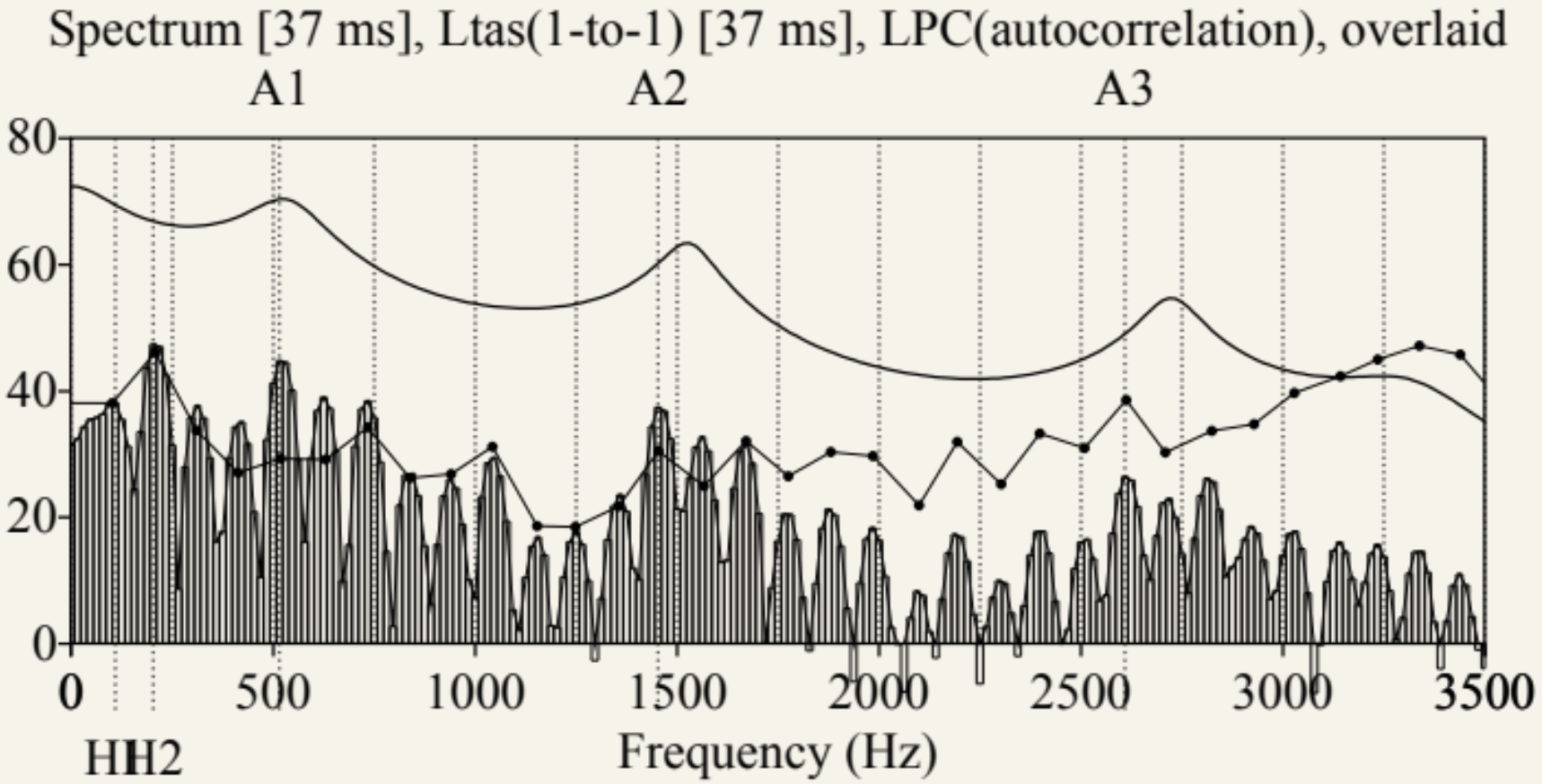
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# Spectral Tilt



# Tone

- Tone is primarily correlated with F0
- But also associated with low-band spectral tilt (H1-H2)

➢ L is breathy

